



Doctor in astrophysics and software engineer, specialized in numerical simulation and performance optimization. Lead developer of simulation and data reconstruction pipelines for the QUBIC experiment, deployed on HPC clusters.

PROFIL

- tomlaclavere@gmail.com
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- Permis B/A2
- TomLaclavère
- Website
- LinkedIn

SKILLS

- Languages: C++ (advanced), Python (advanced), Fortran (intermediate)
- HPC: OpenMP, TBB, MPI (mpi4py), SLURM
↳ familiar with: SYCL, Thrust, Kokkos, CUDA
- Performance: Profiling (perf, Malt, Maqao), Optimisation, Vectorisation
- Tools: Git, CMake, Docker, Linux, CI/CD, unit testing.
- Data Analysis: Scipy, Pandas, sklearn, PyTorch, JAX

SOFT SKILLS

- Rigor and work organisation
- Autonomy and adaptability
- Written and oral communication
- Ability to explain complex technical concepts
- Technical ownership and sense of responsibility

LANGUAGES

- French (native)
- English (professional)
- Japanese (beginner)

HOBBIES

- Sports: Boxing, badminton
- Programming, video games, motorcycling, hiking

TOM LACLAVÈRE

Doctor in Astrophysics - Software Engineer - Numerical Simulation, Scientific Computing & HPC

EXPERIENCE

2023-2026

CNRS |
Université
Paris-Cité |
APC Laboratory

PhD - QUBIC Experiment

- Lead developer of the simulation, reconstruction and data analysis pipeline (qubicsoft, > 30k LoC, > 650 commits).
- Large-scale linear system solving: numerical methods & machine learning.
- Parallelisation (mpi4py, OpenMP) and SLURM deployment (up to 20 nodes, 100-200 GB/run).
- Development of calibration and noise analysis tools for data validation.

2025-

GitHub
Project

HPC / Simulation Project - 3DPhysicsEngine (C++23)

- 3D rigid body simulation engine: custom math library, two-phase collision pipeline, three integrators (Euler, Verlet, RK4)
- Benchmarks: 150+ configurations (solver × time step), convergence, performance and energy drift quantified
- Current phase: CPU optimization (SIMD, OpenMP); roadmap: CUDA/SYCL, MPI

EDUCATION

2023-2026

PhD in Fundamental Physics - Astrophysique

- QUBIC Data Analysis: Realistic astrophysical components reconstruction and atmospheric mitigation using spectral imaging (APC)
- Goal: Constrain Inflation parameters by detecting primordial CMB B-modes, separating astrophysical foregrounds and atmospheric contamination via spectral imaging.

2025

Gray Scott Summer School - HPC (CC-FR/LAPP)

- TBB, OpenMP, OpenMPI, SYCL, CUDA, Thrust
- C++, Fortran, Rust, Python, Julia

2023

Master Nuclei, Particles, Astroparticles & Cosmology (NPAC) - Mention bien (Université Paris-Cité)

2021

Bachelor's in Fundamental Physic - Mention Très bien (Université Paris-Cité)

2018-2020

CPGE, Physics-Chemistry - Admitted to Centrale (Cité Scolaire Bertran-de-Born)

TEACHING

2024 - 2026

CentraleSupélec
Université
Paris-Cité
Ecole
d'Ingénieur
Denis Diderot

Adjunct Teaching Assistant

- Cosmology - 1ère année - Tutorials (35h) - 2026
- Physique Numérique - Master 1 - Tutorials (36h) - 2024 & 2025
- Champs Electromagnétique - 1ère année - Tutorials (24h) - 2024
- Bruits - 2ème année - Lab sessions (24h) - 2024

PUBLICATIONS

- Atmosphere mitigation in Time-Ordered-Data using Spectral Imaging with QUBIC instrument : ongoing
- Spectral Imaging with QUBIC: building frequency maps from Time-Ordered-Data using Bolometric Interferometry ([arXiv:2409.18698](https://arxiv.org/abs/2409.18698))
- Spectral Imaging with QUBIC: building astrophysical components from Time-Ordered-Data using Bolometric Interferometry ([arXiv:2409.18714](https://arxiv.org/abs/2409.18714))
- Neural-Network Map-Making : ongoing
- Contribution to the 2024 Cosmology session of the 58th Rencontres de Moriond: Bolometric interferometry and spectral: a QUBIC overview ([arXiv:2406.15414](https://arxiv.org/abs/2406.15414))

CONFERENCES & TRAININGS

- 59th Rencontres de Moriond, Cosmology session, 2026, Italie, La Thuile
 - Presenter, "Spectral Imaging with QUBIC: Component separation methods using Bolometric Interferometry and application on atmospheric mitigation". (20 min)
- Gray Scott School, 2025, France, Annecy
 - Summer School, "The GRAY SCOTT SCHOOL 2025 - Revolutions will be a deep dive into High Performance Computing, computing optimisation, profiling, and software engineering, to guide you through important topics such as CPU/GPU architectures, Unit Tests, Computing Precision, Memory Allocation and profiling, with modern C++, Rust, Fortran and Python languages, and libraries such as Sycl, EVE, Vulkan, CUDA, Thrust, PyTorch."
- Machine learning in Python with scikit-learn, 2025, France Université Numérique & Inria, online
 - Training session, Data Analysis with scikit-learn
- Groupement De Recherche in Cosmological Physics (GDR CoPhy) Episode 3, 2025, France, Paris, ENS
 - Presenter, "Spectral Imaging with QUBIC: Component separation methods using Bolometric Interferometry". (25 min)
- Scientific Python Training CC-IN2P3, 2025, France, Lyon
 - Training session, Python training focused on High Performance Computing with Python and utilisation of IN2P3 Computational Center (using SLURM)
- Ecole Rodolphe Cledassou, 2024, France, Hendaye
 - Summer School, "L'école Rodolphe Cledassou forme les jeunes chercheurs francophones à la science des futurs grands relevés cosmologiques. Son fonctionnement est assuré par les organismes et instituts français impliqués dans la mission spatiale Euclid".
- 58th Rencontres de Moriond, Cosmology session, 2024, Italie, La Thuile
 - Poster, "Bolometric interferometry and spectral: a QUBIC overview"